

Observation (5)

(homomorphism property for $k[x]/(f(x))$)

$$f(x) = \sum a_i x^i$$

Suppose that we have

a ring homomorphism

$$\lambda: k \rightarrow B$$

(automatically injective)

Then

"extending" λ

$$\left\{ \begin{array}{ccc} k[x]/(f(x)) & \xrightarrow{\quad} & B \\ \uparrow \lambda & \nearrow \lambda & \\ k & & \end{array} \right\}$$

$$\left\{ h \in B : \begin{array}{l} \text{"} f(h) \text{"} = \sum_i \lambda(a_i) h^i = 0 \end{array} \right\}$$

$$\text{ev}_{\lambda, h}: \sum a_i x^i \mapsto \sum \lambda(a_i) h^i$$

$$\text{ev}_{\lambda, h}(f(x)) = 0$$

Pl:

$$k[x]$$

$$\downarrow$$

$$k[x]/(f(x))$$

$$\xrightarrow{\quad \bar{\lambda} \quad}$$

$$B \ni h$$

injective

$$\rightarrow \uparrow \lambda$$

$$k$$

$$\nearrow \lambda$$

$$k$$

$$\longrightarrow k[x]$$

$$\longrightarrow k[x]/(f(x))$$

$$\longrightarrow$$

$$k[x]/(f(x))$$

$$a \mapsto a \mapsto a + (f(x))$$

Rmd. On HW 1, problem 4,
we basically saw the
example where

$$k = \mathbb{F}_p$$

$$f(x) = x^2 + x + 1$$

Slogan: $k[x]/(f(x))$ is the
gateway for zeros of
 $f(x)$

Rmk:

$$k[x]/(f(x)) \cong \left\{ r(x) \in k[x] : \begin{array}{l} \deg r(x) \\ < \deg f(x) \end{array} \right\}$$

$\forall$

$$k \cong \{ r(x) : \deg r(x) = 0 \}$$

Defⁿ

$$k \leq L$$

is a subfield

if k & L
are fields &
 k inherits $+$, $\cdot$, 0 , 1
from L

e.g. $\mathbb{C} \leq \mathbb{R}$

Application of Observation

Defⁿ $k \leq L_1$
 $\leq L_2$: two field extensions of k

- $(R \subseteq \mathbb{C})$
 $\cdot \frac{k[x]}{(f(x))}$
 $f(x)$ irreducible
 L is a field extension of k

$Emb_k(L_1, L_2) = \left\{ \begin{array}{l} \lambda: L_1 \rightarrow L_2 : \text{ring hom.} \\ \text{s.t. } \forall a \in k \leq L_1 \\ \lambda(a) = a \end{array} \right\}$
 $\lambda(a) = a$
 $\in k \leq L_2$

"
Embeddings of
 L_1 in L_2 as
 field extensions
 of k

Rmk: "Embeddings"
 is justified because
 any such λ is
 automatically injective
 & so puts L_1 inside
 of L_2 without loss
 of information

Application : $k \leq \frac{k[x]}{(f(x))}$, $f(x)$ irreducible
 $\leq L$: field extension
 Then
 $Emb_k\left(\frac{k[x]}{(f(x))}, L\right) = \{ \lambda \in L : \lambda \text{ is a ring hom. s.t. } \lambda(a) = a \}$

$$\begin{array}{ccc}
 & \parallel & \text{ring homomorphism} \\
 & \downarrow & \\
 \left\{ \begin{array}{ccc} k[x]/(f(x)) & \xrightarrow{\psi} & L \\ \uparrow \varphi & & \\ R & \hookrightarrow & \end{array} \right\} & \hookrightarrow & \{f(t)=0\}
 \end{array}$$

$$\text{Emb}_R(k[x]/(f(x)), L) = \{t \in L : f(t) = 0\}$$

Example:

$$R = \mathbb{Q}, \quad f(x) = x^2 + 1 \in \mathbb{Q}[x]$$

$\wedge$

L : any field ext. of $\mathbb{Q}$

$$\text{Emb}_{\mathbb{Q}}(\mathbb{Q}[x]/(x^2+1), L) \cong \{t \in L : \underset{=0}{t^2+1}\}$$

$$\Rightarrow \bullet \text{Emb}_{\mathbb{Q}}(\mathbb{Q}[x]/(x^2+1), \mathbb{R}) = \emptyset$$

$$\begin{aligned}
 \bullet \text{Emb}_{\mathbb{Q}}(\mathbb{Q}[x]/(x^2+1), \mathbb{C}) \\
 \cong \{i, -i\}
 \end{aligned}$$

Finite fields

k : finite field

Dichotomy
 k : field

Then either

Characteristic 0 $\rightarrow$

$2 \leq k$

or

Characteristic p

$\exists$ prime p
s.t. $p \cdot 1_k = 0$

Observation (1)

k has characteristic

p for some prime p

$\Leftrightarrow \mathbb{F}_p \leq k$: field extension
 $a \mapsto a \cdot 1_k$

Proposition: $k^\times$ is cyclic

Observation (2): Let $\alpha \in k^\times$ be a generator. Then

$$k = \{0, 1, \alpha, \dots, \alpha^{m-2}\}$$

Where $m = |k|$

Observation (2)' $\alpha \in k^\times$ a generator

$$\begin{array}{ccc} \mathbb{F}_p[x] & \xrightarrow{\text{ev}_\alpha} & k \\ \sum a_i x^i & \longmapsto & \sum a_i \alpha^i \end{array}$$

Then ev_α is surjective
 $\text{ev}_\alpha(x^j) = \alpha^j$

$$\begin{array}{ccc} \mathbb{F}_p[x] & & \\ \downarrow & \searrow \text{ev}_\alpha & \\ \mathbb{F}_p[x] / \ker \text{ev}_\alpha & \xrightarrow[\text{ev}_\alpha]{\cong} & k \end{array}$$

$\Rightarrow$